

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Performed mechanical assembly, electrical connections, and robot calibrations for new semiconductor manufacturing equipment from integration to final inspection, resolving ~5-7 technical issues per installation cycle.
- Followed detailed installation checklists and schematics for complex systems, ensuring adherence to procedures and flagging non-conformances with engineering teams.
- Collaborated with senior engineers and operations teams to test equipment functionality using diagnostic techniques and provided daily status reports on progress and challenges.
- Participated in continuous improvement initiatives focused on reducing installation cycle times, contributing to a documented reduction of approximately 15% in average assembly duration for specific modules.
- Managed installation project timelines and quality requirements for multiple concurrent equipment deployments, ensuring all tasks met stringent technical specifications and customer readiness.
- Responded to factory-reported issues by engaging with cross-functional engineering teams and participating in escalation calls to resolve critical installation blockers.

SKILLS

Engineering & Installation – Mechanical Assembly, Electrical Connections, Robot Calibration, Equipment Installation, Troubleshooting, Technical Documentation, Diagnostic Techniques, Preventative Maintenance, Corrective Maintenance

Software & Tools – MS Office Suite (Excel, PowerPoint), Jira, Confluence, Git

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker

Programming & Scripting – Python, Java, SQL

Databases – PostgreSQL, MySQL, Oracle

CI/CD & Operations – Jenkins, Splunk, Dynatrace

Analytics & Reporting – Power BI, Tableau, DAX, Power Query

PROJECTS

Semiconductor Equipment Installation & Validation (HCL Technologies)

- Executed end-to-end installation of complex semiconductor fabrication equipment, including mechanical, electrical, and vacuum system integration, troubleshooting ~30% of encountered issues independently.
- Developed and refined installation procedures and schematics for a new line of wafer processing tools, improving clarity and reducing misinterpretation by 20% across the installation team.
- Assisted in equipment testing and calibration, ensuring all parameters met design specifications and documenting test results for client handover and internal quality assurance.

Automated Production Reporting & Dashboarding (HCL Technologies)

- Engineered Python scripts to automate the extraction, transformation, and loading (ETL) of operational data from production systems into a PostgreSQL data warehouse.
- Developed interactive Power BI dashboards to visualize key performance indicators (KPIs) related to equipment uptime, throughput, and error rates for operations management.
- Integrated data pipelines with Jenkins for scheduled execution, providing daily operational reports that enabled faster identification of system anomalies and reduced manual reporting effort by 8 hours per week.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)